Physics III
031
03/08/2022 08:30 AM – 10:00 AM

NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

ADVANCED LEVEL NATIONAL EXAMINATIONS, 2021-2022

SUBJECT: PHYSICS

PAPER III: PRACTICAL

COMBINATIONS:

PHYSICS –CHEMISTRY- MATHEMATICS **(PCM)**

PHYSICS –CHEMISTRY- BIOLOGY **(PCB)**

MATHEMATICS- PHYSICS-GEOGRAPHY **(MPG)**

MATHEMATICS-PHYSICS- COMPUTER SCIENCE **(MPC)**

DURATION: 1H 30 MIN

INSTRUCTIONS:

1) Write your names and index number on the answer booklet as written on your registration form and **DO NOT** write your names and index number on additional answer sheets of paper if provided.
2) Do not open this question paper until you are told to do so.
3) This paper consists of **one** compulsory question. **(40 marks)**
4) You may use non-programmable calculator and mathematical set where appropriate.
5) All answers should be written in the answer booklet provided.
6) Use a **blue** or **black** pen and pencil for drawing.
7) Tie your ray-trace sheet into your answer booklet. It will be marked

ATTEMPT ALL SUB-QUESTIONS OF THIS QUESTION (40 marks)

In this experiment, you are required to investigate reflection of light using a plane mirror.

Materials required:

1 plane mirror with a support that holds it vertically

1 white sheet of paper A4

1 protractor

1 transparent ruler 30 cm or 15 cm long

1 pencil with rubber

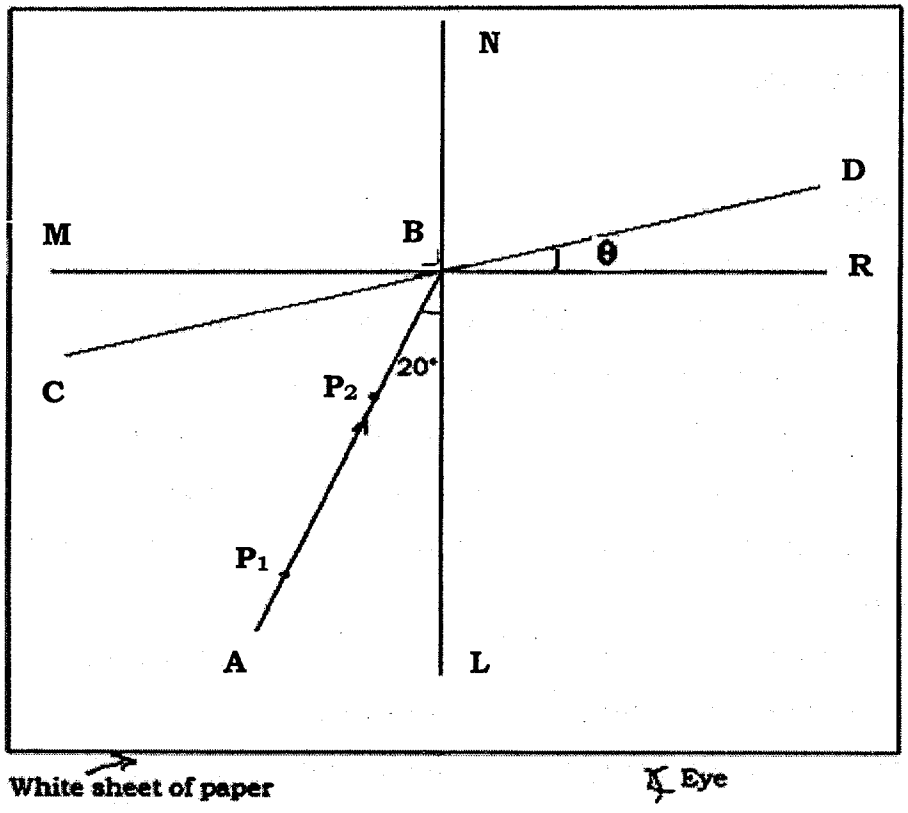
4 drawing pins/office pins

4 optical pins

1 soft board/drawing board

Procedures

Carry out the following instructions referring to the figure below drawn not to scale.



a) Fix the white sheet of paper on the soft board using 4 drawing pins.
b) Draw a line 15 cm long near the middle of the white sheet of paper.

Label the line **MR**. Draw a normal **NL** to this line that passes through its centre **B**.

c) Draw a line from **B** at an angle of incidence i = 20.0° to the normal **NL**, below **MR** and to the left of **NL**. Label the end of this line **A**.

The normal NL and the line AB remain the same during this experiment.

d) Place the reflecting face of the plane mirror vertically on the line **MR**

The additional line CD shows a second plane mirror position.

e) Place two optical pins $\mathbf{P_1}$ and $\mathbf{P_2}$ on the line **AB** a suitable distance apart.

f) View the images of optical pins $\mathbf{P_1}$ and $\mathbf{P_2}$ from the direction indicated by the eye in the figure above. Place two optical pins $\mathbf{P_3}$ and P_4 some distance apart so that the optical pins $\mathbf{P_3}$ and $\mathbf{P_4}$ and the images of $\mathbf{P_1}$ and $\mathbf{P_2}$ all appear exactly on behind the other.

g) Remove optical pins $\mathbf{P_3}$ and $\mathbf{P_4}$ and the plane mirror. Draw the line joining the positions of $\mathbf{P_3}$ and $\mathbf{P_4}$. Extend the line until it meets NL

h) Measure and record the angle **β** between NL and the line joining the positions of $\mathbf{P_3}$ and $\mathbf{P_4}$.

At this stage the angle θ between the plane mirror and the line MR is 0° and β= β_0 .This is the first reading

i) Remove the optical pins $\mathbf{P_1}$ and $\mathbf{P_2}$. Draw the lines at angles θ=5.0°, 10.0°,15.0°,20.0° and 25.0° to **MR**. One line **CD** at a certain angle **θ** is shown in the figure where **θ** is the angle between the new position of the mirror and the line **MR** . Repeat the procedures (e) to (h) **placing the mirror on each of the new lines in turn** so that you obtain six sets of readings.

Questions

1) Tabulate your results in a suitable table including the angles θ, β and $\beta - \beta_0$. **(12 marks)**

2) Draw a graph of the angle β (y-axis) against θ (x-axis)
Draw a line of best fit. **(11 marks)**

3) From the indication of the slope shown on the graph, determine the slope **S** of this graph. **(2 marks)**

4) Suggest the relationship between β and θ. You may express the relationship in words or as an equation. **(1 mark)**

5) Suggest the relationship between $\beta - \beta_0$ and θ and conclude. **(2 marks)**

6) Does your conclusion correspond to the aim of this experiment? Defend your idea. **(3 marks)**

7) State two precautions you would take in this experiment to improve accuracy.

(2 marks)

8) Some errors made when you carried out the experiment are unknown but the absolute error can be estimated.
What is the maximum possible error committed by using your protractor? **(1 mark)**

9) **Insert** and **attach** the sheet of paper used into the answer booklet and submit both to avoid being treated as a cheat. **(6 marks)**